Worksheet W6 · W-Seminar "Time"

Design your study

Before you collect a single number: find the kind of data that can really answer your question – and talk your plan through.

1 My question – version 3

make version 2 from W5 testable

spark → question → working thesis → **testable claim** → thesis

Question: *How does the tempo of background music change how long a 3-minute wait feels?*

Testable claim (hypothesis): *Fast music (140 bpm): lower estimates in minutes than slow music (60 bpm).*

Changed since W5 – and why: *made measurable: fixed wait, two tempos, estimate in minutes*

2 Which quality criterion is at risk?

Objectivity: same result whoever runs or scores it · **R**eliability: same result again, little noise · **V**alidity: measures what you claim. Tick the criterion that is hit first.

	O	R	V
a) You mark students' open answers yourself – without a marking key.	☒	☐	☐
b) One single question ("Are you stressed?") is your whole measure of stress.	☐	☒	☐
c) A speed-reading task is used to show how well students *understand* a novel.	☐	☐	☒
d) Group A waits before lunch, group B after lunch.	☐	☐	☒

3 Six ways to get data – brainstorm for your question

Think 4 min Fill in what each type could give you – even the odd ones; a weird idea often leads to a good one. Pair 4 min Swap sheets. Your partner ticks the option that convinces them most and writes one question in the margin.

type	what exactly could I collect?	+ / –	best?
Survey what people say	*ask 100 students how long typical waits feel – with and without music*	*+ many people; – memory, vague answers*	☐
Test what people can do	*(hardly fits: perception is not a skill)*	*– wrong construct*	☐
Experiment what causes what	*3-minute wait with fast / slow music, estimate in minutes*	*+ shows cause; – rooms, time, permission*	☒
Interview how and why	*talk to 6–8 students about waits that felt long or short – and why*	*+ reasons, new ideas; – few people, no cause*	☐
Coding texts count categories	*how song lyrics or novels describe waiting*	*+ no participants; – a different question*	☐
Existing data others' statistics	*results of published music-and-waiting studies*	*+ large samples; – not my own data*	☐

My partner's question: *Would people notice the music at all if they are on their phones?*

4 Talk it through: partner interview

Pair 2 × 6 min Your partner asks the questions and notes your answers in keywords. Then swap roles. Don't defend – explain; "I don't know yet" is a fine answer and goes on your to-do list.

partner asks	answers (keywords)
1 What exactly do you measure – and how does it become a number or category?	*estimated waiting time in minutes, written on a card*
2 Who takes part, how many, and how are they chosen?	*Year 10–11, 2 × 30, groups drawn by lot*
3 What else could explain your result?	*room, time of day, knowing the song, phones*
4 Which result would show that your thesis is wrong?	*no difference – or slow music feels shorter*
5 What could go wrong in practice – what is your plan B?	*no permission → online version at home*

My partner's best tip: *add a question afterwards: how fast did the music feel?*

5 My design

decide – you can still change it after the pilot

☐ survey ☐ test ☒ experiment ☐ interview ☐ coding texts ☐ existing data ☐ mixed: _____

Idea → rule (operationalisation): *"time flies" → estimate of a 3-min wait in minutes*

Variables / items / categories: *IV tempo (fast / slow), DV estimate; same room, volume*

Sample (who, n, how chosen): *2 × 30 students, Year 10–11, random assignment*

Procedure, step by step:

1. same room and script 2. 3 minutes of music 3. estimate on a card 4. tempo rating

6 Secure the three criteria in my study

Share 1 min each Present your design to another pair. They name the biggest risk they see – add it to the table.

	biggest risk	what I do about it
Objectivity	*I explain things differently*	*written script, same for all*
Reliability	*rough timing*	*stopwatch, exactly 180 s*
Validity	*confounds, weak manipulation*	*same room and time, tempo rating*

Before I start ☒ asked my teacher ☐ school permission / parents' consent ☒ anonymous codes ☒ pilot with *5* people